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Palaeomagnetism of NW Scotland syenites in relation to local and regional tectonics

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Summary. A detailed thermal demagnetization study has been carried out on a number of syenitic intrusions from north of the Great Glen Fault in Scotland. Multicomponent remanences of normal and reversed polarity have been recognized. These define a series of palaeomagnetic poles from which an Ordovician to Silurian apparent polar wander path for northern Scotland is deduced. Comparison of this apparent polar wander path with the established path for Britain throws new light on the sequence of thrusting and magmatic activity in the NW Highlands and precludes large-scale ($\lesssim 1500$ km) movement along the Great Glen Fault later than the Ordovician.

The data emphasize previously published discrepancies between the apparent polar wander paths of Britain, Baltica and Laurentia and are consistent with the suggestion that Britain was among a set of intracratonic microplates which underwent complicated, distinct movements throughout the Caledonian orogeny.

1 Introduction

Lower Palaeozoic palaeomagnetic data for Scotland are confined to the Southern Uplands and the Highlands between the Highland Boundary and Great Glen Faults while Proterozoic data are confined to the Lewisian foreland region. Hence the data distribution is inadequate to make a full palaeomagnetic assessment of the amount of relative translation and rotation within the region during the Caledonian orogeny.

Recent studies in the Alpine-Mediterranean region (e.g. Channell & Horvàth 1976; Edel, Montigy & Thuizat 1981) and the North American western cordillera (e.g. Irving 1979; Coney, Jones & Monger 1980) have shown that translations and rotations of both large and small blocks within orogenic belts are to be anticipated in addition to the large-scale crustal shortening brought about by the collision of major plates.

In the Caledonides, geological evidence (McKerrow & Cocks 1976; Thirlwall 1981) supports the existence of an ocean, the Iapetus, between the Southern Uplands and the Lake District during the Cambrian and Ordovician which finally closed in late Silurian or early Devonian time. Briden, Morris & Piper (1973) attempted to assess the overall amount of closure across the Caledonides by comparison of Ordovician palaeomagnetic data from England and Wales with that from the Younger Gabbros of Aberdeenshire and estimated

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post-Ordovician closure as approximately 1000 +/− 800 km while expressing doubts as to whether this estimate was significantly different from zero. More recently Deutsch (1981, 1982) has suggested post-mid-Ordovician closure of 330 +/− 2200 km within Ireland. These estimates are unsatisfactory because of the large uncertainties associated with the figures. More importantly, this postulated closure across the Iapetus ocean is small compared with the discrepancies between these British Ordovician-Silurian data and the contemporary North American data on conventional north Atlantic reconstructions (e.g. Bullard, Everett & Smith 1965) suggesting that an even larger tectonic discontinuity existed between Britain and the North American craton margin. Van der Voo & Scotese (1982) have argued that large (~ 2000 km) post-Caledonian movement along the Great Glen Fault could account for these discrepancies. Further, in the Devonian the British data are not only discordant with North American data, but also with those from the Baltic region.

As a contribution to the enhancement of the database, which is a prerequisite to the solution of these problems, intrusive rocks in the thrust belt marginal to the deformed Caledonides and adjacent to the stable foreland have been studied palaeomagnetically and the results are reported in this paper. They have all been regarded as of Ordovician age although direct evidence of this is only available for some of them.

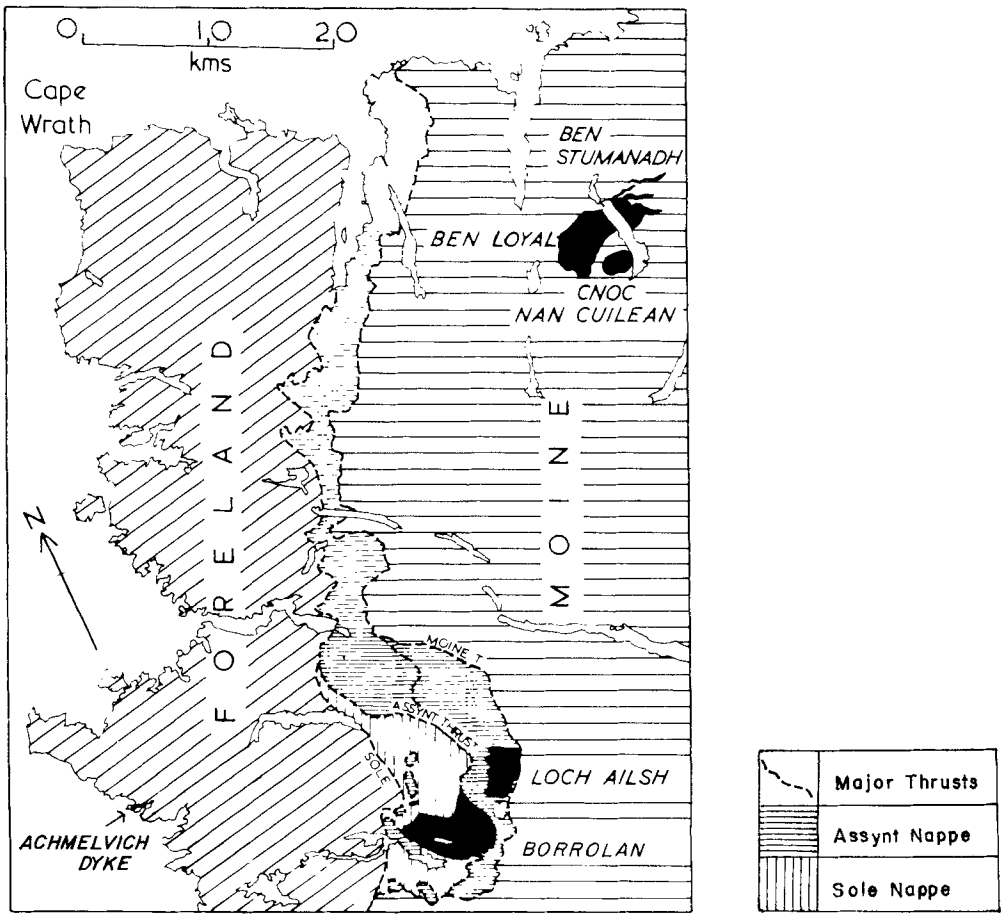


Figure 1. Distribution of the major alkaline intrusions in NW Scotland. The major nappes are also shown (after Parsons 1972).

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The main sampling areas of this study are Assynt and Ben Loyal (Fig. 1). The Assynt district is centred on a series of thrusted Cambrian sediments between the Lewisian foreland to the west and the Moine schists to the east. The Assynt igneous suite comprises two major intrusions, the Borrolan and Loch Ailsh centres (Fig. 1), and a complex of dykes and sills. The Ben Loyal igneous complex lies some 30 km north of Assynt and intrudes Moine metasediments. Additionally, a small collection was made from the Achmelvich dyke intruding Lewisian gneisses on the stable Caledonian foreland some 20 km west of Assynt.

The multicomponent remanences found in these rocks will be discussed in relation to the timing and scale of local thrusting as well as in relation to the larger scale plate movements associated with Caledonian orogenesis.

2 Geology

Almost all the successful palaeomagnetic studies on the Scottish Lower Palaeozoic have been made on dioritic, andesitic, basaltic and ultrabasic igneous rocks. Many of the more acid rocks are palaeomagnetically unstable, e.g. Newer Granites of Garabal Hill (Briden 1970) and Criffel-Dalbeattie (S. J. Kneen, unpublished data). The subjects of the present study are mainly alkaline intrusive rocks, although composition varies considerably within and between the complexes. Chemical and isotopic age data suggest a genetic relationship to the Newer Granite suite (Turnell 1982).

The position of the intrusions on the different thrust sheets can be seen from the map (Fig. 1). The Sole, Ben More, Glencoul and Assynt thrusts form an imbricate series which youngs to the west (Elliot & Johnson 1980). The Moine Thrust overrides the other nappes from the east and it may be that this difference in geometry of the Moine nappe is related to the Assynt intrusions having already been emplaced in the imbricate pile.

The Borrolan complex (Fig. 2) comprises an outer suite and an inner suite separated by

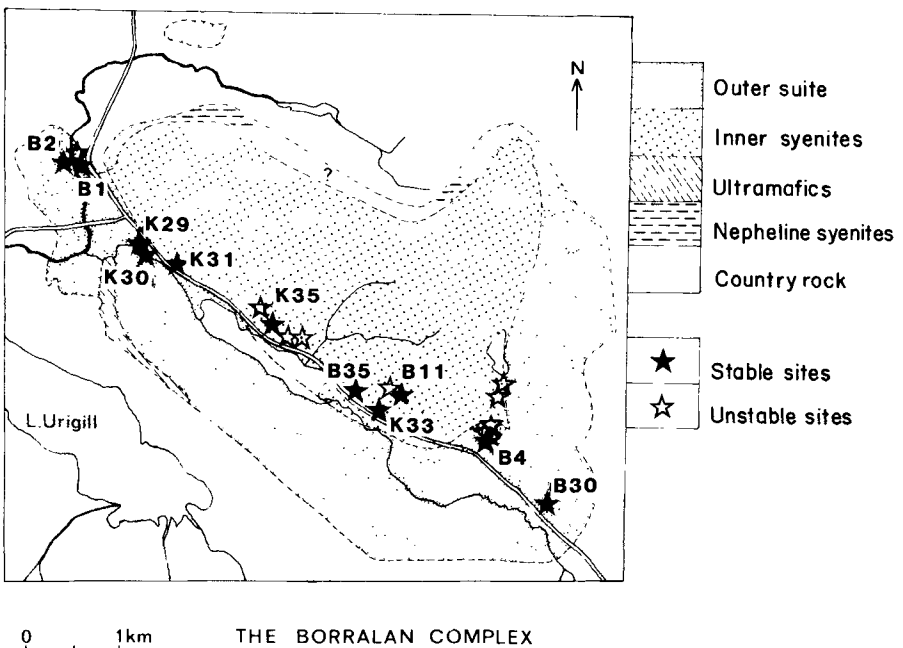


Figure 2. Sketch map of the Borrolan Complex showing the distribution of sampling sites between the two major phases (after Woolley 1973).

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an intrusive junction. The outer suite is subdivided into ledmorite-type exposed in the north-west and possibly younger pseudoleucite-type exposed in the south-east. These form a conformable sheeted complex which was subsequently punched through by the plug-like body of inner syenite. Exposure is poor, especially of the outer phases, for which sampling was limited to the river sections, a quarry and isolated exposures in the peaty lowland. A total of 21 sites were drilled, of which 10 were in the inner phase (Fig. 2).

The Loch Ailsh intrusion consists of a variety of broadly uniform leuco-syenites which are well exposed, and a total of six sites were drilled.

Lamprophyric dykes and sills exposed along the road and along the shore of Loch Assynt, north of Borrolan, were sampled at nine sites. The Achmelvich dyke was sampled at the coast near Lochinver. It is petrographically similar to ledmorite suggesting a common magmatic origin (Sabine 1953).

The Ben Loyal syenite complex comprises three distinct centres of varying grain size and composition. The coarse rock of Ben Loyal itself was sampled at one site, eight sites were drilled in the finer-grained, most mafic Cnoc nan Cuilean intrusion, including two sites in cross-cutting aplite dykes.

3 Method

Routinely, individual specimens from each site were thermally demagnetized to 700°C mainly in steps of 50°C, and in steps of 10–20°C between 450 and 600°C. Another specimen from each site was af demagnetized in steps of 5–10 mT. Only after a definite pattern

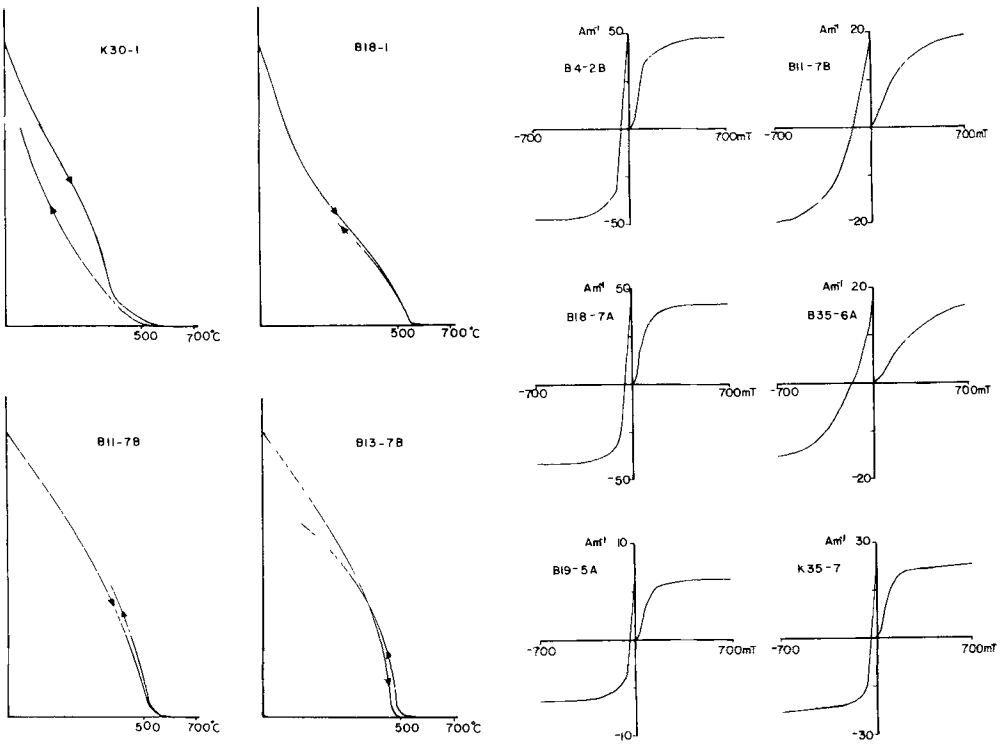


Figure 3. Typical thermomagnetic and hysteresis curves from the Assynt collection. Curie points indicate magnetite and titanomagnetite. Unstable specimens, e.g. B18-1, commonly show a large drop in Js at low temperatures. Haematite is revealed only by the hysteresis curves, e.g. B11, B35 and K35. The vertical scale for the thermomagnetic curves, Js, is arbitrary.

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of behaviour had emerged for each site was the rate of experimentation increased by using fewer, appropriately chosen temperature and field intervals.

In general, straight line segments on orthogonal projections of demagnetization data define single components of NRM. Although it is often difficult to determine exactly which points on a demagnetization trajectory 'belong' to a straight line segment, some decision is necessary in order to make 'best estimates' of the direction of magnetization components and of their blocking temperature (or coercivity) ranges. In this study, minimization of the 95 per cent confidence angle $\Delta\alpha_{95}$ on vector differences ($\Delta\gamma_{95}$ of Briden & Arthur 1981) was used as the criterion.

Hysteresis and thermomagnetic curves were generated for specimens from each formation to provide additional information about the remanence carriers (Fig. 3).

4 Results

NRM intensities of all specimens were in the range 10 to 500×10^{-3} A m^{-1} with directions close to that of the present field. Koenigsberger ratios, *Qn*, peaked around 0.2 for material from the Borrolan outer phase but were more evenly distributed between 0 and 1.0 for the rest of the specimen collection. These results indicate the widespread occurrence of coarse-grained magnetite.

Af demagnetization was not generally successful in isolating ancient components of magnetization. Most specimens showed erratic directional changes at field strengths greater than 20 mT. This behaviour is attributed to the high proportion of viscous remanence carriers in the rocks which pick up stray fields either in the demagnetizer or during measurement. Upon thermal treatment, however, a high proportion of specimens showed smooth demagnetization trends from which one or more ancient components of magnetization could be determined.

In situ remanence directions identified in the magnetically stable specimens are listed with their blocking temperature ranges, in Tables 1–6 and their directions are illustrated in Figs 4 and 5. No bedding corrections were applied because there is no evidence that post-cooling tilt affected the major intrusions. The remanence directions are labelled N (normal) = upward northerly, R (reversed) = downward southerly, or B (downward northerly) in the analysis described below. Question marks indicate anomalous components. The data are grouped according to formation and, for the Borrolan outer phase, are further subdivided into rock-units. Hence, individual specimens are given unit weight in the Fisher (1953) analysis and calculation of mean poles.

4.1 BORROLAN OUTER PHASE

Most specimens revealed two components of magnetization (Table 1 and Fig. 5a): a negative north-northwesterly component (interpreted as primary, normal magnetization, N) carried by magnetite and haematite with blocking temperatures in the range 500–690°C, and a south-southeasterly positive component (R) of lower blocking temperature (approximately 200–500°C) which is interpreted as a later PTRM acquired in a reversed magnetic field (e.g. K30-6, Fig. 4a, b). Reversed components were rare at sites B2 and K29. Some of the reversed directions had steeper inclinations than the rest of the collection. In the absence of any other diagnostic feature they have been included with all the other data from the ledmorites in calculating a single palaeomagnetic pole. However, it is possible that these steeper directions are composites due to overlap of blocking temperature ranges of shallow N and R components. It is also possible that they are younger than the other components in these rocks and correlated with reversed magnetization in the inner syenite (see later).

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Table 1. NRM components in the Borrolan Ledmorite and contact zone identified by blocking temperature range.

SPEC.	TEMP. RANGE °C	COMPONENT D	COMPONENT I	$\Delta\alpha_{95}$	COMPT.
LEDMORITE					
B1-1A	172-493	130.0	77.8	4.1	R
	493-583	21.5	-15.6	7.9	N
B1-2	186-496	181.5	67.9	10.7	R
	496-638	18.9	-27.7	26.6	N
B1-3B	186-496	186.8	75.1	5.4	R
B1-4B	206-502	174.9	40.5	17.4	R
	502-576	17.8	-38.4	26.8	N
B1-6A	206-400	144.9	31.5	7.5	R
	502-576	4.1	-34.8	4.6	N
B1-6B	206-400	160.4	28.4	19.2	R
	502-590	8.9	-35.4	5.1	N
B2-1	525-650	338.5	-50.6	3.2	N
B2-2	525-650	342.2	-43.3	1.9	N
B2-3	502-576	324.0	-40.8	12.5	N
B2-4	502-576	7.2	-60.3	4.2	N
B2-6B	542-602	341.3	-34.8	2.9	N
B2-7	200-406	194.6	48.7	26.0	R
	502-610	333.8	-44.3	3.6	N
K29-1	409-586	334.6	-22.9	10.0	N
K29-2	485-624	322.0	-38.9	9.8	N
K29-4	206-400	167.0	59.8	10.5	R
	400-576	354.8	-34.4	11.2	N
K29-5	361-586	354.2	-57.0	4.0	N
K29-6	491-618	315.6	-45.3	8.4	N
K30-1A	311-566	340.5	-25.7	9.9	N
K30-2	206-400	166.6	12.7	21.3	R
	502-576	343.0	-17.9	5.4	N
K30-3B	545-624	316.2	-1.1	11.2	N
K30-4B	559-618	345.1	-28.3	27.9	N
K30-5B	261-409	149.8	15.1	32.3	R
	409-586	340.2	-21.0	7.1	N
K30-6	200-400	166.0	14.9	15.9	R
	400-565	350.0	-21.7	9.3	N
CONTACT ZONE					
K31-1	150-261	166.7	22.5	37.4	R
K31-3	494-578	127.9	27.0	11.9	R
K31-4	477-630	154.5	26.4	8.0	R
K31-5	525-609	158.4	40.8	8.1	R

The blocking temperatures and ranges quoted refer to the data points between which the removed vector has been calculated. They do not, therefore, necessarily cover exactly the total blocking temperature range for each component. *D*, *I* represent declination (°E of N) and inclination (positive downward) of removed vectors during progressive demagnetization. $\Delta\alpha_{95}$ is the cone of 95 per cent confidence about that direction (Briden & Arthur 1981). The notation N, R, B of remanence components is explained in the text.

Site K31 is the only one from the contact zone between the outer and inner phases. Its NRM is unique among the syenite in revealing only a south-easterly positive component with a high magnetite blocking temperature. This direction is characteristic of the outer phase (Fig. 5a) although the rock specimens are mineralogically similar to the material of the inner intrusion. However, the dark, magnetite rich bands observed in thin section may be part of the mafic ledmorite and dominate the high temperature remanence. Therefore, the results were included in the ledmorite mean.

The south-eastern pseudoleucites of the Borrolan outer phase were sampled at sites B4, in a quarry and B30. Specimens from B4 contained both reversed and normal components with blocking temperatures approximately in the ranges 200–500°C and 500–580°C

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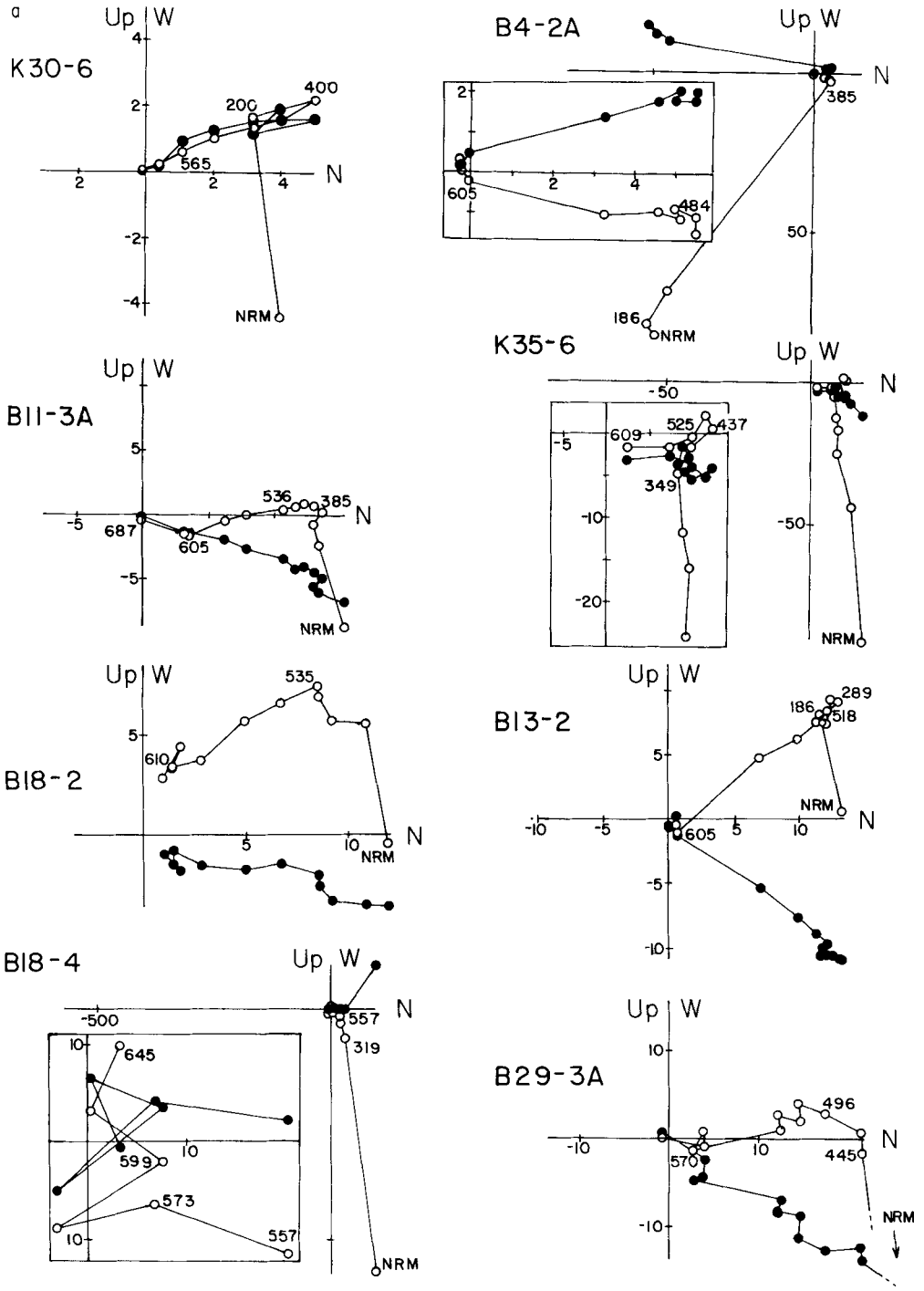


Figure 4. (a) Orthogonal projections of thermal demagnetization behaviour. Comparison of B18-2 and B18-4 provides an example of within site variability. For B18-4, B4-2A and K35-6 details of the high temperature behaviour are inset. The axes are marked in units of $\times 10^3$ A m^{-1}. Open symbols represent points on the vertical plane and solid symbols represent points on the horizontal plane. Temperatures, in °C, mark the points chosen for calculating the removed vectors listed in Tables 1–6. (b) Stereographic projections of thermal demagnetization data corresponding to the orthogonal projections in (a). Open symbols represent upper hemisphere directions and closed symbols represent lower hemisphere directions.

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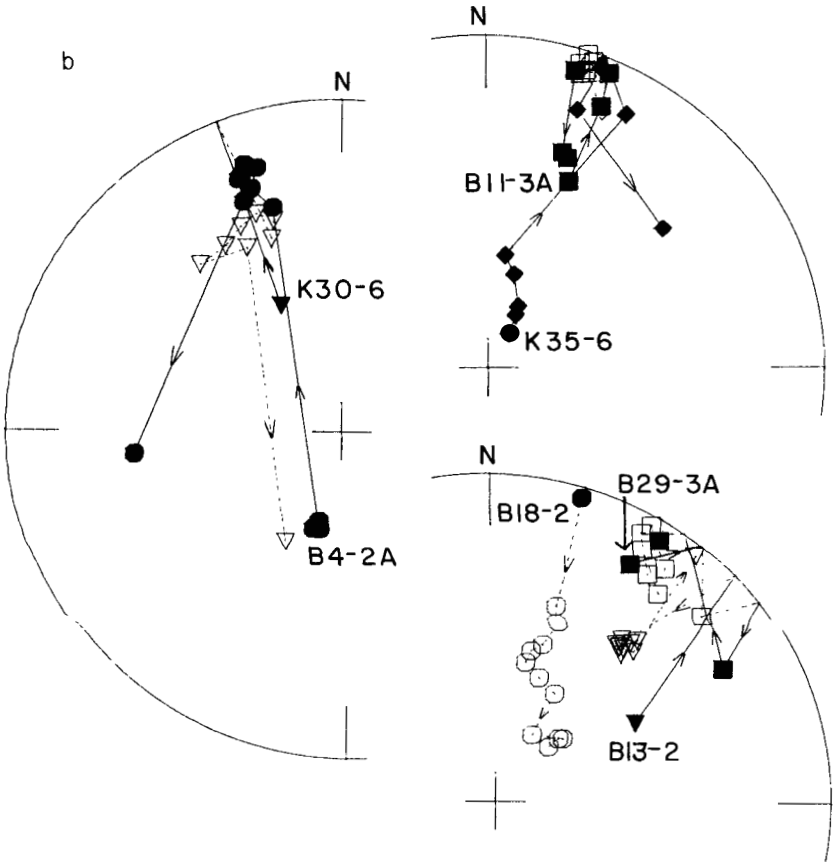


Figure 4 – *continued*

respectively (Fig. 4a, b and Table 2). Material from site B30 is highly altered although not obviously sheared. Over half the specimens appeared to be totally remagnetized by lightning, but the remainder contained stable ancient magnetizations, with blocking temperatures appropriate to haematite, similar to the high blocking temperature directions in B4.

4.2 BORROLAN INNER PHASE (SYENITE)

The directional scatter among primary magnetization directions in the three sites B32, B35 and K35 is low (Fig. 5b and Table 3), and the inner phase mean, which has a north-easterly negative direction, is significantly different from any results from the outer intrusion (Fig. 5f and Table 7). Blocking temperature ranges between 500 and 580°C for most specimens indicate that they contain a single magnetite ancient remanence, referred to as N magnetization henceforth. Only one specimen, K35-6, contained a southerly R component with blocking temperature around 400°C, interpreted as a later PTRM acquired in a reversed field (Fig. 4a). Specimens B35-6B and B32-6 contain evidence of a north-easterly positive direction, B, carried by haematite, and magnetite of low blocking temperature, respectively (Table 3).

The remaining sites from the Borrolan inner phase, B11 and K33, were drilled in material of slightly different composition. This mineralogical variation is attributed to differentiation (Shand 1910; Woolley 1973). Nevertheless, most specimens contained a

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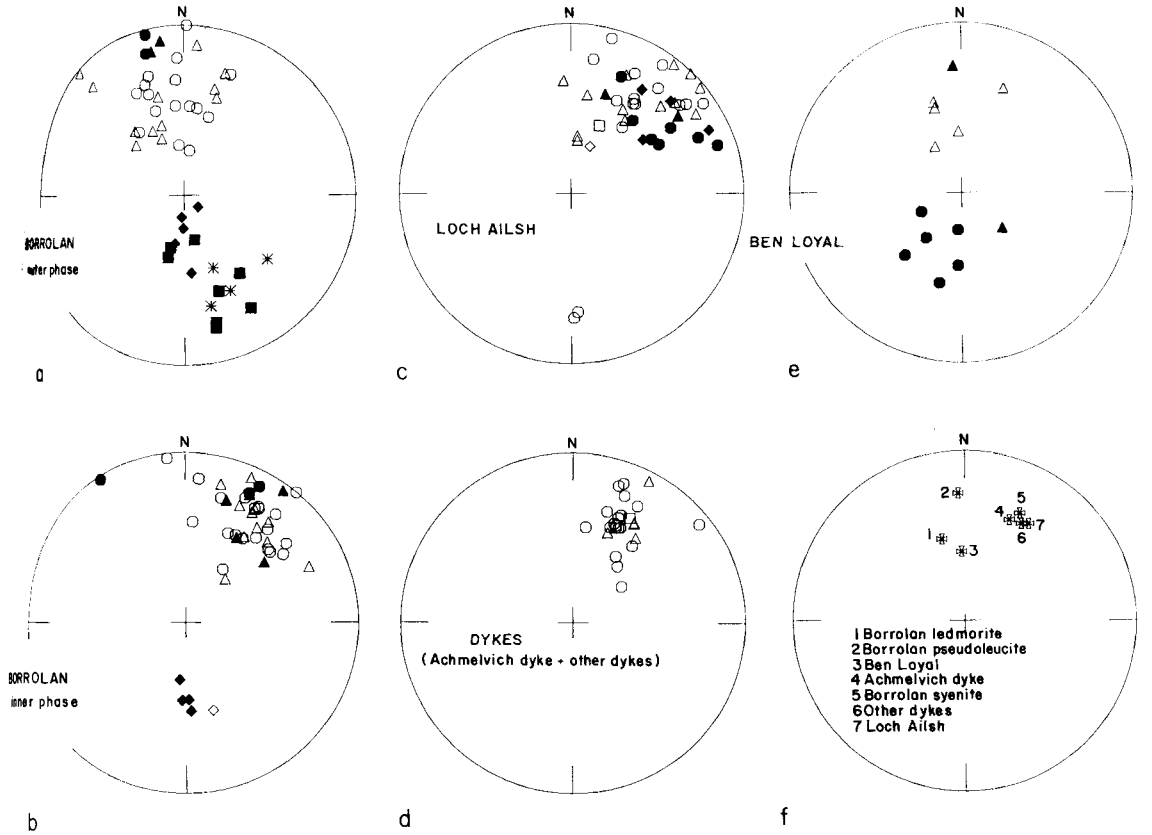


Figure 5. (a–e) Stereographic projections for each rock unit illustrating the removed vectors listed in Tables 1–6. The different symbols broadly subdivide the data according to maximum blocking temperature as follows: squares 400°C; diamonds 400–500°C; circles 500–600°C; triangles > 600°C. In (a), data from site K31 are plotted as stars. (f) An equal-angle stereogram showing the mean directions from each rock unit (see Table 7).

Table 2. NRM components in the Borrolan pseudoleucite identified by blocking temperature range.

SPEC.	TEMP. RANGE °C	COMPONENT D	I	$\Delta\alpha_{95}$	COMPT
B4-1B	173-521	191.2	57.8	3.4	R
	521-631	318.2	-8.9	4.3	N
B4-2A	186-484	193.1	53.4	4.4	R
	484-605	346.8	8.3	5.1	N
B4-3	522-566	344.5	8.5	16.9	N
B4-4	481-600	357.2	-12.3	7.6	N
B4-10	200-412	194.6	54.7	9.9	R
	412-588	346.1	1.7	14.3	N
B4-11	529-591	351.1	4.9	8.8	N
B30-1A	487-634	16.5	-24.0	4.2	N
B30-3	522-577	1.1	-0.1	4.4	N
B30-4B	487-634	19.0	-15.9	5.3	N
B30-7	570-606	5.2	-7.1	24.7	N

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Table 3. NRM components in the Borrolan inner phase (syenite) identified by blocking temperature range.

SPEC.	TEMP. RANGE °C	COMPONENT D	I	α_{95}	COMPT
B11-1	433-565	26.2	9.6	15.1	B
B11-2	474-610	50.0	-24.8	7.0	N
	610-704	36.7	-20.5	13.2	N
B11-3A	536-605	15.9	-15.5	8.3	N
	605-687	18.4	15.7	7.9	B
B11-5	393-602	25.5	-12.0	11.0	N
	602-704	26.5	10.0	13.0	B
B11-6	439-626	24.7	-15.9	3.5	N
B11-7A	484-642	31.0	-16.1	5.4	N
	642-687	30.7	29.4	6.6	B
B11-8	426-580	353.7	-1.7	9.3	N
	580-712	52.2	30.0	13.0	B
B11-9	400-555	25.1	-29.9	13.8	N
B11-10	200-437	176.7	35.2	10.6	R
	439-588	29.5	-29.0	13.1	N
	588-674	14.4	-10.1	9.8	N
K33-1	548-590	328.6	1.2	5.6	N
K33-2A	354-451	177.8	41.3	15.5	R
	508-592	4.1	-28.9	5.1	N
K33-3	243-467	182.7	40.9	12.1	R
	467-624	31.0	-14.0	14.3	N
K33-4	626-674	15.5	-13.6	12.2	N
K33-5	165-475	167.9	-33.6	28.6	?
	475-614	42.3	-52.0	7.6	N
B32-2	534-589	40.2	-0.1	10.2	N
B32-3	556-725	24.3	-4.0	7.5	N
B32-6	445-534	28.4	5.6	13.0	B
	534-613	45.2	-27.6	21.6	N
B35-1	502-570	39.6	-11.2	20.8	N
B35-2A	453-600	39.3	-15.6	5..	N
B35-3	511-587	41.2	-18.3	6.8	N
B35-4	550-572	34.0	-28.7	11.9	N
B35-6B	446-600	39.5	-24.7	7.1	N
	600-647	36.3	2.4	18.3	B
B35-7B	520-611	31.3	-14.3	8.3	N
B35-8	539-641	34.0	-27.5	12.2	N
K35-1	385-540	32.1	-12.7	9.0	N
K35-2	481-584	48.3	-24.6	16.7	N
K35-3	485-576	35.0	-48.5	35.8	N
K35-4	396-532	51.6	-16.7	10.1	N
K35-5	475-559	32.7	-13.2	21.3	N
K35-6	349-437	186.0	52.6	5.9	R
	525-609	5.4	-9.4	17.0	N
K35-8	477-559	54.8	-20.7	10.3	N

north-easterly negative component, N, with blocking temperatures in the range 450–600°C as well as a southerly component with lower blocking temperatures (<400°C) which is correlated with the reversed magnetization R in site K35 (Table 3 and Fig. 5b). This R magnetization has southerly declination and is not perfectly antiparallel to N. It could be due to slightly overlapping blocking temperatures of the normal and reversed magnetizations, but we have no direct evidence for this. It could be of significantly different age from N. However, so few data are involved that we have calculated their statistics together and any distortion due to this procedure is small. The site means of the N component from B11 and K33 have slightly lower declination than the other Borrolan syenites but the difference is not significant.

In five out of nine specimens from B11 there is convincing evidence of a north-easterly positive remanence, B, with blocking temperature extending into the haematite magnetization range (Fig. 4a), similar to that seen in B35-6. The origin of the B direction is uncertain and these data may be composite vectors resulting from the overlapping blocking temperatures of N and R. The presence of haematite is confirmed in the hysteresis curves for specimens from B11-7 and B35-6 (Fig. 3).

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Table 4. NRM components in Loch Ailsh identified by blocking temperature range.

SPEC.	TEMP. RANGE °C	COMPONENT D	I	95	COMPT
B13-1	399-533	56.4	20.5	25.2	B
	553-586	37.3	-37.7	5.7	N
B13-2	518-605	36.1	-33.9	4.9	N
B13-4	289-536	60.5	29.4	24.6	B
	536-642	44.7	-0.7	8.7	N
B13-5B	301-534	55.8	31.1	15.9	B
B13-6	244-505	46.9	13.2	22.7	B
		34.5	-24.7	5.6	N
B13-9	298-506	66.0	7.2	13.0	B
	506-590	39.4	-12.8	6.1	N
B13-10	526-570	26.2	-27.4	3.9	N
	570-603	24.8	-15.0	6.7	N
B14-1	399-540	55.5	-4.6	24.0	N
	540-593	45.5	-17.6	6.7	N
B14-2B	267-520	66.0	12.2	11.4	B
	520-607	57.2	-8.4	5.5	N
B14-3	398-583	71.6	6.4	21.4	B
	533-630	48.5	-12.2	9.7	N
B14-4	520-603	31.6	-29.7	55.7	N
B14-5	516-586	33.6	-22.6	23.5	N
B14-6	398-586	35.5	-24.2	5.6	N
B15-1A	270-504	34.2	17.2	11.1	B
	504-613	38.3	-2.1	3.1	N
B15-2B	288-486	52.8	34.8	19.9	B
	486-602	35.2	-4.7	5.5	N
B15-3	395-538	40.0	31.4	23.8	B
	538-638	50.5	-1.6	6.1	N
B15-4	413-594	51.9	-9.3	8.2	N
B15-5	398-586	177.1	-20.2	7.7	?
B15-6	224-586	179.0	-17.7	10.6	?
B16-1	553-630	19.0	26.6	20.5	B
B16-2B	180-430	21.8	-56.7	7.9	N
B16-4	224-398	22.6	-43.3	20.3	N
B16-6B	270-576	27.7	-13.0	13.5	N
B17-2B	572-610	6.8	-55.0	18.9	N
B17-3B	586-686	6.3	-52.4	22.4	N
B17-4A	504-573	14.7	-3.4	23.5	N
B17-6	530-640	4.5	-12.8	14.8	N
B17-7	546-586	50.6	-11.5	17.4	N
B18-1	533-630	356.4	-22.8	33.0	N
B18-2	535-610	7.5	-24.2	11.2	N
B18-4	433-592	23.2	16.6	9.0	B
B18-6	501-604	53.9	14.9	23.4	B

4.3 LOCH AILSH

NRMs of the Loch Ailsh rocks showed considerable variation (Table 4). Well-defined, multi-component remanences were identified in sites B13 to B15, but a large proportion of specimens from sites B16 and B18 contained only viscous magnetizations (Fig. 4a). Single component ancient remanences were determined from a few specimens in these sites but the directions were associated with high errors ($\Delta\alpha_{95} > 20$). The hysteresis curve for B18-7 (Fig. 3) is characteristic of material containing a high proportion of multi-domain titanomagnetite. The rocks contained a stable north-easterly negative magnetization (component N) similar to the Borrolan inner phase (Fig. 5c). Specimens with the lowest proportion of viscous components also contained definite evidence of a lower blocking temperature positive north-easterly component, B, of similar direction to that seen in some sites from the inner phase of the Borrolan complex. It has the blocking temperature characteristics of a later PTRM but because of the anomalous direction, satisfactory interpretation requires that it be a product of two components, one of which may be the ancient NE upper hemisphere direction (N). The second component, which may be younger because of the lower magnetite blocking temperatures, may have been reversed, or is possibly a recent viscous magnetization. In support of the reversed option it is noted that while there were hints of reversed components in several specimens especially in the low temperature steps (e.g.

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186–289°C in B13-2, Fig. 4a) either the inclinations were steep and it was impossible to resolve a reversed positive component from a present field direction, or the stable blocking temperature range was too short for a reliable estimate. If this is the correct interpretation, it is surprising that, from the complete Loch Ailsh collection, there are no clear examples of reversed magnetizations in contrast to results from both Borrolan phases.

Anomalous but stable southerly negative magnetizations are contained in two specimens from site B15.

4.4 ACHMELVICH DYKE

Although only two localities were sampled in the Achmelvich dyke, the definition of ancient magnetizations is good, as shown by the low $\Delta\alpha_{95}$ values in Table 5.

Table 5. NRM components in dykes identified by blocking temperature range.

SPEC.	TEMP. RANGE °C	COMPONENT D	COMPONENT I	α_{95}	COMPT
ACHMELVICH DYKE					
B19-1	322-590	22.3	-25.4	4.8	N
B19-2	444-606	23.9	-27.2	6.0	N
B19-3	424-553	23.0	-22.2	7.4	N
B19-4	396-604	21.0	-27.9	6.6	N
B19-5B	456-616	20.7	-30.9	3.2	N
B19-7	452-564	21.9	-12.3	13.4	N
B20-1B	336-574	26.2	-26.2	11.7	N
B20-3	322-591	25.3	-26.6	5.7	N
B20-4	297-353	28.2	-20.1	9.4	N
B20-5B	312-551	23.8	-20.7	8.0	N
B20-6B	328-600	23.4	-26.0	4.2	N
OTHER DYKES					
B29-1	476-570	37.2	-31.0	16.4	N
B29-2	492-566	19.7	-8.0	17.1	N
B29-3A	496-570	28.6	-14.0	14.4	N
B29-5A	545-669	31.1	-21.1	13.5	N
B29-6B	539-572	53.1	-51.1	16.4	N
B29-7B	501-583	32.5	-40.5	16.7	N
B29-8B	515-592	38.0	-44.5	17.1	N
B29-9B	514-572	17.3	-9.4	24.1	N
K32-1A	556-594	16.9	-21.3	17.4	N
K32-1B	550-618	31.3	-21.5	16.9	N
K32-2	527-587	7.5	-31.1	18.5	N
K32-3	545-624	36.0	-26.8	11.0	N
K32-4A	428-580	51.5	-4.0	32.8	N
K32-5B	568-609	27.8	-3.3	20.6	N

Hence the palaeomagnetic pole (Table 7) is associated with a low α_{95}. Because the dyke is less than 6 m wide it is likely to have cooled rapidly through the blocking temperature of the titanomagnetic remanence carrier deduced from thermomagnetic and IRM experiments (Fig. 3). The data used to calculate the palaeomagnetic pole, then, may not average out secular variation. The blocking temperatures are dominantly in the range 500–580°C with only one example of a low blocking temperature component (200–400°C) in B20-4 which is one of the few specimens containing very little viscous remanence.

4.5 OTHER DYKES

Stable specimens from the lamprophyre suite were limited to sites K32 and B29. They were considered separately from the Achmelvich dyke because of their different location and coarser texture. The $\Delta\alpha_{95}$ values quoted for the removed magnetization components (Table 5)

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are high owing to viscous components which affected the measurements throughout the demagnetization. The definition of the removed vector for each specimen was improved by making a large number of measurements throughout the stability range of the ancient component (Fig. 4a).

4.6 BEN LOYAL

From the Ben Loyal collection, only the leucocratic Ben Loyal syenite (site B42) and the aplite dykes cutting Cnoc nan Cuilean (sites B37 and B38) were found to contain stable, ancient remanences (Table 6). The directions have declinations intermediate between the

Table 6. NRM components in Ben Loyal identified by blocking temperature range.

SPEC.	TEMP. RANGE °C	COMPONENT D	I	$\Delta\alpha_{95}$	COMPT
CNOC NAN CUILEAN					
B37-3	615-688	330.5	-55.5	11.6	N
B37-5B	533-581	244.1	60.9	11.2	R
B37-6B	519-577	186.2	65.0	6.6	R
B37-7B	581-688	343.2	-32 0	14.0	N
B38-3	577-688	22.0	-23.3	22.3	N
B38-5	581-615	357.2	-50.4	20.0	N
B38-6	577-688	356.2	16.6	22.4	N
B38-2B	567-620	342.8	-35.1	15.5	N
BEN LOYAL					
B42-1	501-602	217.9	52.7	4.4	R
B42-2	501-602	221.9	37.2	9.9	R
B42-5	501-568	182.9	43.4	9.6	R
B42-6B	518-602	193.9	32.6	7 5	R
B42-7	501-675	131.4	55.3	4.1	R

two Borrolan phases (Fig. 5e). The specimens from Ben Loyal itself revealed components with mostly magnetite blocking temperatures and reversed polarity. Of the aplite dykes, specimens from site B38 revealed normal magnetizations with blocking temperatures appropriate to haematite whereas from site B37, reversed components were removed below 580°C and normal components were removed between 580 and 680°C.

5 Interpretation

Fig. 6(a) is a plot of the palaeomagnetic south poles determined from each rock unit. By comparison with the established lower Palaeozoic APWP for Britain (e.g. Piper 1979; Briden & Duff 1981) the magnetizations in these rocks, and perhaps the magmatic activity itself, range in age from upper Ordovician to at least late Silurian, and the following order for the sequence of intrusions is inferred: Borrolan ledmorites, Borrolan pseudoleucites and Ben Loyal, and finally the dykes, the Borrolan syenite and the Loch Ailsh pluton.

The pole positions thus indicate tectonic rotation and/or continental drift in the Ordovician and Silurian. The rate of this movement is uncertain because the precise age range is not well determined, but it could have involved phases of rapid movement.

The 430 Ma age determined for the Borrolan outer phase using lead isotopes (van Breemen, Aftalion & Johnson 1979) correlates well with the palaeomagnetic pole. Unfortunately, there are no isotopic data for the other formations, but various authors have published ideas on the sequence of intrusive events from detailed petrology and mapping. Woolley (1973) suggested that the pseudoleucites were derived from the same magma as

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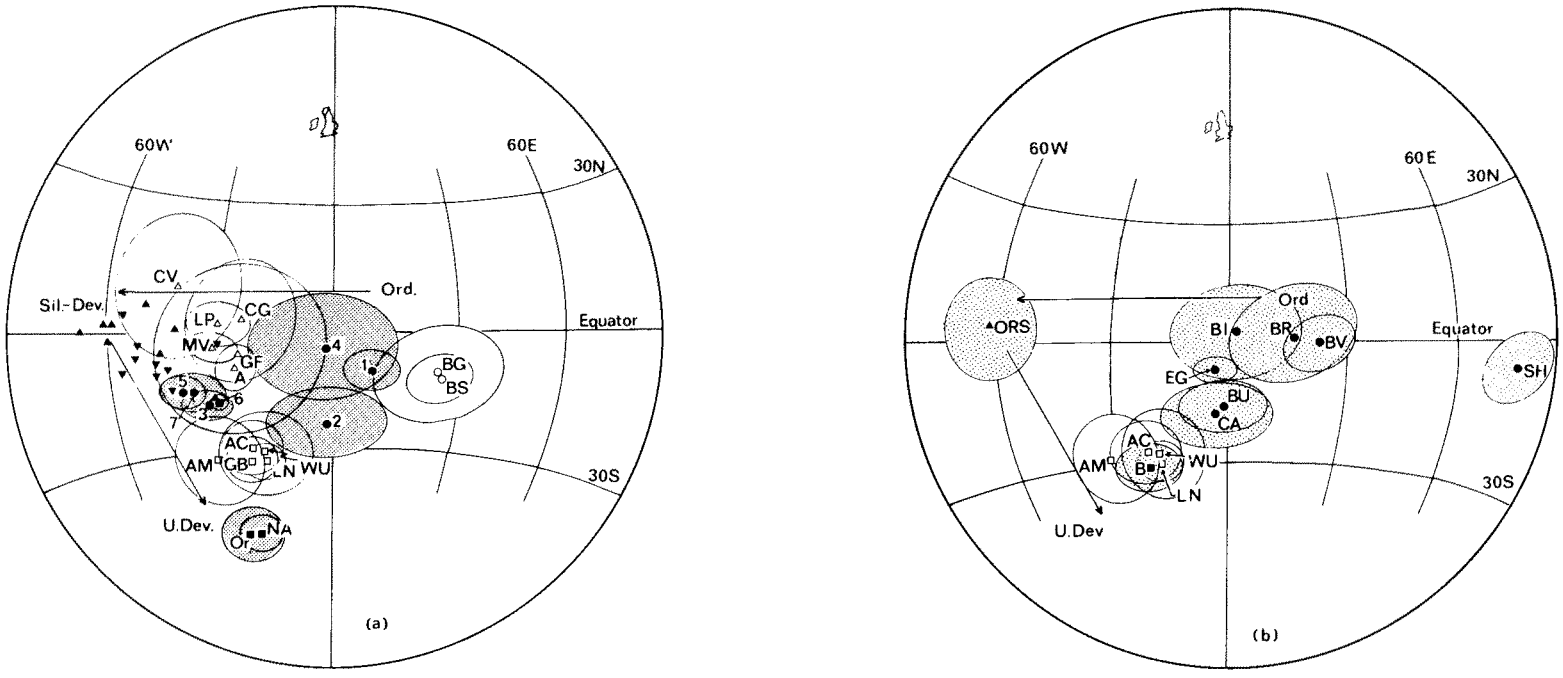


Figure 6. (a) Palaeomagnetic poles of this study compared with other relevant Lower and mid-Palaeozoic poles. Data from north of the Great Glen Fault are shown by solid symbols. Those of this study are shown by ●. They are numbered as in Table 7 and their ovals of 95 per cent confidence are shaded. It is argued in the text that the magnetization ages of these rocks range from Ordovician to Silurian. Those from individual sites (see footnote below) in the Moines are shown by triangles, upright ▲ for intermediate blocking temperatures and inverted ▼ for high blocking temperature (Watts 1982); their ovals of confidence are omitted for clarity. Those claimed by Van der Voo & Scotese (1982) to be representative of the Middle and Upper Devonian field are illustrated by ■ with circles of 95 per cent confidence and denoted Or for Orcadian Basin and NA for North American data corrected for opening of the Atlantic (Bullard *et al.* 1965). Data from south of the Great Glen Fault are shown by open symbols, with their 95 per cent confidence regions. Open circles ○ denote Ordovician data from Scotland – BG, BS = Ballantrae Gabbro and Serpentinite. Open triangle △ denotes Silurian (to possibly Lower Devonian) data from Scotland – MV, LP 'Old Red Sandstone' lavas of the Midland Valley and Lorne Plateau; A and GF the Arrochar and Garabal Hill-Glen Fyne Newer Granite complexes; CV and CG the Cheviot Volcanics and Granite. (Data from further south in Britain are not used because correction for closure of the Iapetus Ocean might be necessary; see Fig. 6b.) Squares □ denote the mean Middle and Upper Devonian poles calculated by Van der Voo & Scotese (1982) for Great Britain south of the Great Glen Fault (GB), the Armorican massif (AM), Leningrad area (LN), Western Urals (WU) and the Acadia region of New England and the Canadian Maritimes (AC) corrected for Atlantic opening. (Lambert equal area projection.) (Note: Results from one locality of Watts 1982 which fall within the area of European Devonian poles are omitted for clarity.) (b) Lower Palaeozoic and Devonian palaeomagnetic data from England and Wales with Middle and Upper Devonian data from other parts of Laurasia. British data are denoted as follows: (Ordovician ●, SH = Shelve volcanics; BV = Borrowdale Volcanic Group; BR = Breidden Hills; BI = Builth Intrusives; BU = Builth Volcanics; CA = Carrock Fell Gabbro; EG = Eycott Group); ORS ▲ = Lower Old Red Sandstone of England and Wales; B ■ = Upper Old Red Sandstone/Lower Carboniferous of Bristol District. The Devonian data from elsewhere are denoted as in (a).

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Table 7. Palaeomagnetic summary statistics.

	INTRUSION	LOCALITY Lat.N	Long.W	DIRECTION D	I	S	k	R_s	$\Delta\alpha_{95}$	PALAEOMAGNETIC SOUTH POLE Lat.S	Long.E	δp	δm
1	BORROLAN LEDMORITE (N & R)	58.1	-5.0	344.4	-37.2	37	12.6	34.2	6.8	10.0	9.7	4.0	7.3
2	BORROLAN PSEUDO-LEUCITE (N & R)	58.0	-4.9	357.2	-16.3	13	7.9	11.5	15.6	23.6	358.0	8.3	16.2
3	BORROLAN SYENITE (N & R)	58.0	-4.9	26.9	-19.4	36	11.0	32.8	7.5	18.4	326.-	4.1	7.4
4	LOCH LOYAL (N & R)	58.4	-4.4	357.2	-45.9	14	7.4	12.2	15.6	4.2	358.0	13.4	20.1
5	LOCH AILSH (N)	58.1	-4.9	33.1	-21.6	29	13.6	26.0	7.6	15.6	321.3	4.3	6.1
6	ACHMELVICH DYKE (N)	58.2	-5.3	23.6	-24.2	11	218.0	11.0	3.0	16.5	330.6	1.8	3.2
7	OTHER DYKES (N)	58.2	-5.0	29.9	-23.3	14	17.6	13.3	9.7	15.5	324.5	5.6	10.4

S is number of (specimen) vectors used in the calculation. Rs is the unit vector sum. k is Fisher's (1953) precision estimate. $\Delta\alpha_{95}$ is the semi-angle of the cone of 95 per cent confidence about the mean direction. δp, δm are the semi-axes of the oval of 95 per cent confidence about the mean pole.

the ledmorites, but it is difficult to determine from geological evidence whether the chemical differentiation occurred before the intrusion of the separate groups or after emplacement, i.e. *in situ*. The implication that the igneous activity spanned (say) 20 Myr, though the inferred ages for poles 5–7 in Table 7 and Fig. 6 are rather younger than most geologists would have anticipated, is not unreasonable. However, it is surprising that the intrusion of the syenite left no obvious palaeomagnetic imprint on outer phase rocks close to the contact; if such an imprint does exist, then it is contained in the R magnetizations in the outer phase and has erratic geographic distribution.

Field evidence indicates that, while the main thrust movements in the Assynt area preceded the Borrolan intrusion, small displacements recurred, producing minor shearing and faulting (Woolley 1971). These had no discernible effect on the palaeomagnetic record.

The proximity of poles from the dykes and the syenite indicates that the injection of the dykes, which are restricted to the Sole nappe, coincided with the intrusion of the Borrolan syenite in spite of their ledmorite affinities. Also, the close agreement between the Borrolan syenite pole and that derived from the Loch Ailsh pluton suggests that these were closely coeval. In a detailed comparative study of the geochemistry of rocks from Borrolan and Loch Ailsh Parsons (1972) revealed differences which show that they cannot have been co-magmatic, although they may have been broadly contemporaneous. In a later paper, Parsons (1979) suggested that the Loch Ailsh pluton pre-dated the Assynt thrust, however, the palaeomagnetic results do not contain any evidence of a magnetization which could be older than that associated with the Borrolan intrusion. It is possible (but unlikely) that the palaeomagnetic direction corresponds to a later, post-thrust remagnetization rather than to the original cooling of the body. The Loch Ailsh data can be interpreted, as here, without invoking any post-cooling movement on the Moine thrust.

6 Conclusions

The palaeomagnetic data in this paper are derived from igneous intrusions of a limited age range. With the aid of radiometric age data and some evidence on the sequence of magmatic activity, the spread of poles can be interpreted as an Apparent Polar Wander Path for late Ordovician to Silurian times. To this can be added the poles from the Moines which

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Watts (1982) associated with late Caledonian orogeny. Together, these are the poles denoted by solid symbols in Fig. 6(a). Also shown in that figure are the published poles from Scotland south of the Great Glen.

Van der Voo & Scotese (1982) have argued for ~2000 km of post-Devonian sinistral movement on the Great Glen on the basis of 20° discrepancy between mid and Upper Devonian poles from either side of it (the solid and open squares in Fig. 6a). Unless virtually the same amount of dextral motion had previously occurred rapidly, close to the Lower/Middle Devonian boundary, this same 20° discrepancy should be evident in the Lower Palaeozoic data astride the Great Glen. On the basis of visual inspection of Fig. 6(a) this does not appear to be so. Both regions seem to have moved in concert, and their apw paths could well be identical.

However, the paths from the two regions would appear to match even better if the poles from south of the Great Glen were displaced ~14° southward (on the present grid of Fig. 6a). This would reconcile the more precise poles from Silurian (? Lower Devonian) volcanic and Newer Granites with poles 3, 5, 6, 7 of the present study which we have already argued in Section 5 are likely to relate to a rather younger age than the usually assumed Ordovician age of those rocks.

The Ballantrae poles would move to the latitude of pole 2 (Fig. 6a), but without necessarily becoming contradictory with poles 1, 2 or 4 because of various uncertainties associated with them. Thus on the basis of the data presented here, post-Lower Palaeozoic sinistral movement on the Great Glen Fault of up to 1500 km cannot be ruled out – the amount depending on the position of the Euler pole which describes the relative rotation. Clearly this matter is not settled, although we shall treat it in more detail elsewhere with the aid of further currently unpublished data (Briden *et al.*, in preparation). The Lower Palaeozoic apw paths from Scotland (Fig. 6a) and England/Wales (Fig. 6b) are not so distinct as to enable unequivocal recognition of the Iapetus ocean between these two areas or of estimation of its width. However, the Scottish Ordovician poles (Fig. 6a) all lie south of the equator while three of the six English poles (Fig. 6b) lie in very low northerly latitudes. This could be indicative of an Iapetus ocean ~1000 km wide in the Ordovician. Closure of this ocean could have continued through the Silurian or early Devonian though the palaeomagnetic data are too sparse and imprecise to confirm or deny this. Continued convergence would be consistent with the idea that the Newer Granites and Old Red Sandstone volcanics are associated with continuing subduction of oceanic crust (Thirlwall 1981).

The poles summarized in Fig. 6(a) reinforce the long-recognized discrepancies with the North American and Baltic continents. If relative motion along the Great Glen Fault accounts for no more than part of the discrepancy with North America, then further transcurrent faults remain to be identified. This would be analogous to the situation elsewhere along the North American craton margin where discrepancies between displaced terrains and the craton appear to be taken up on several faults (Kent 1982). While major discontinuities such as the Moine Thrust and the Flannan Thrust [recently discovered by seismic reflection profiling off NW Scotland (Smythe *et al.* 1982)] could have been the sites of some transcurrent movement in early stages of their evolution, the evidence presented in this paper strongly suggests that the majority of the 20° discrepancy between British and North American data has to be taken offshore of NW Britain and of the North American seaboard because the data from Assynt (west of the Moine Thrust) are of more British than American affinity. It follows from this that straightforward matching of the Scottish Lewisian foreland with the Greenland Archaean as on the Bullard *et al.* (1965) or similar reconstructions is unlikely to be valid.

The implication of the discrepancy between the British and Baltic data for major

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Palaeozoic discontinuity in the North Sea (Briden *et al.* 1973; Van der Voo & Scotese 1982) remains enigmatic pending unequivocal data of critical ages from Scandinavia. However, a picture does seem to be emerging in which Britain, and perhaps neighbouring parts of the Caledonian belt, comprised a set of intracratonic microplates and terrains which underwent complicated rotations and transcurrent displacement along the belt in the late stages of Caledonian orogeny.

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Appendix: site locations

Rock unit	Site	Grid reference
Borrolan	B1	NC 242 131
Outer phase	B2	NC 243 132
	K29	NC 248 121
	K30	NC 249 120
	B4	NC 288 096
	B30	NC 293 090
	K31	NC 253 118
Borrolan	B11	NC 277 102
Inner phase	K33	NC 274 102
	B32	NC 261 114
	B35	NC 272 103
	K35	NC 263 113
Loch Ailsh	B13	NC 324 126
	B14	NC 327 127
	B15	NC 328 129
	B16	NC 332 133
	B17	NC 334 176
	B18	NC 335 143
Achmelvich	B19	NC 059 243
Dyke	B20	NC 053 235
Other dykes	B29	NC 252 187
	K32	NC 249 127
Ben Loyal	B37	NC 611 472
	B38	NC 609 472
	B42	NC 612 498

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